

1. Problem

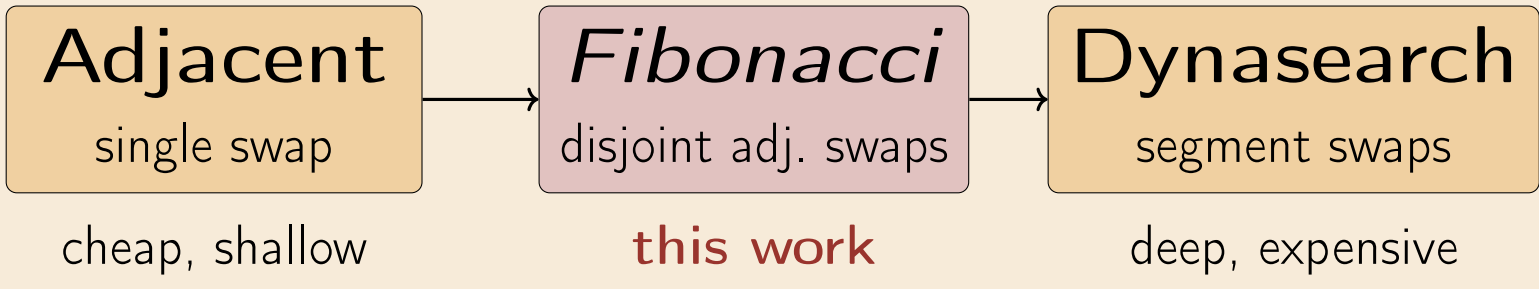
Permutation Flow Shop Problem (PFSP). n jobs processed in the same order on m machines; minimize the makespan

$$C_{\max}(\pi) = C_{m,n}(\pi).$$

NP-hard for $m \geq 3$ — metaheuristics are the standard tool.

Question. Inside Tabu Search (TS) and Simulated Annealing (SA), which *neighborhood structure* performs best under a fixed wall-clock time budget?

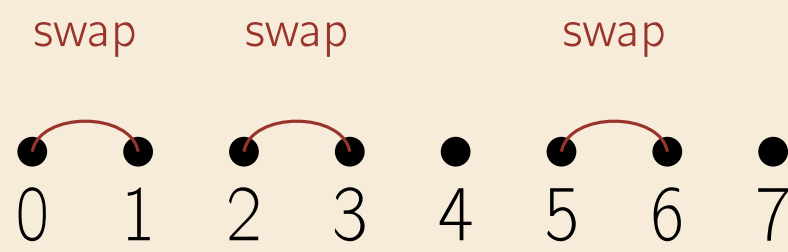
2. Neighborhood structures



Composite Disjoint-Adjacent (*Fibonacci*). A move is a subset $S \subseteq \{1, \dots, n-1\}$ of adjacent-swap positions with the **non-overlap** constraint

$$i \in S \implies i+1 \notin S.$$

All selected swaps $\pi_i \leftrightarrow \pi_{i+1}$ are applied *simultaneously*.



Counting. The number of valid subsets in a permutation of length n equals the Fibonacci number F_{n+1} — hence the nickname.

3. Algorithm & cost

Per iteration:

- Recompute $C_{\max}(\pi^{(i)})$ for each $i = 1, \dots, n-1$ — each $O(mn)$.
- One-pass DP picks the optimal non-overlapping subset:
 $dp[i] = \min(dp[i+1], \Delta_i + dp[i+2]).$
- Apply selected swaps; fall back to the least positive Δ_i if DP picks none.

Neighborhood	Candidates	Per iter.
Adjacent	$n - 1$ swaps	$O(mn)$
<i>Fibonacci</i>	$n - 1$ swaps + DP	$O(mn^2)$
Dynasearch (rec.)	$O(n^2)$ segments	$O(mn^3)$

Fibonacci explores an **exponentially-sized** neighborhood (F_{n+1} valid subsets) at **quadratic** per-iteration cost — the sweet spot between cheap Adjacent and expensive Dynasearch.

4. Metaheuristics

Tabu Search (TS). Tabu list keyed by primitive moves or composite signatures; tenure $\tau_{\text{tabu}} = 10$; aspiration on global-best improvement.

Simulated Annealing (SA). Multiplicative cooling $T \leftarrow \alpha T$ with floor; **time-based reheating**: if no improvement for $\tau_{\text{stagn}} = 500$ ms and $T < T_0$, set $T \leftarrow \min(T_0, r \cdot T)$, $r = 1.5$.

Same neighborhoods plugged into both frameworks — any performance gap is attributable to the neighborhood, not the framework.

5. Experimental setup

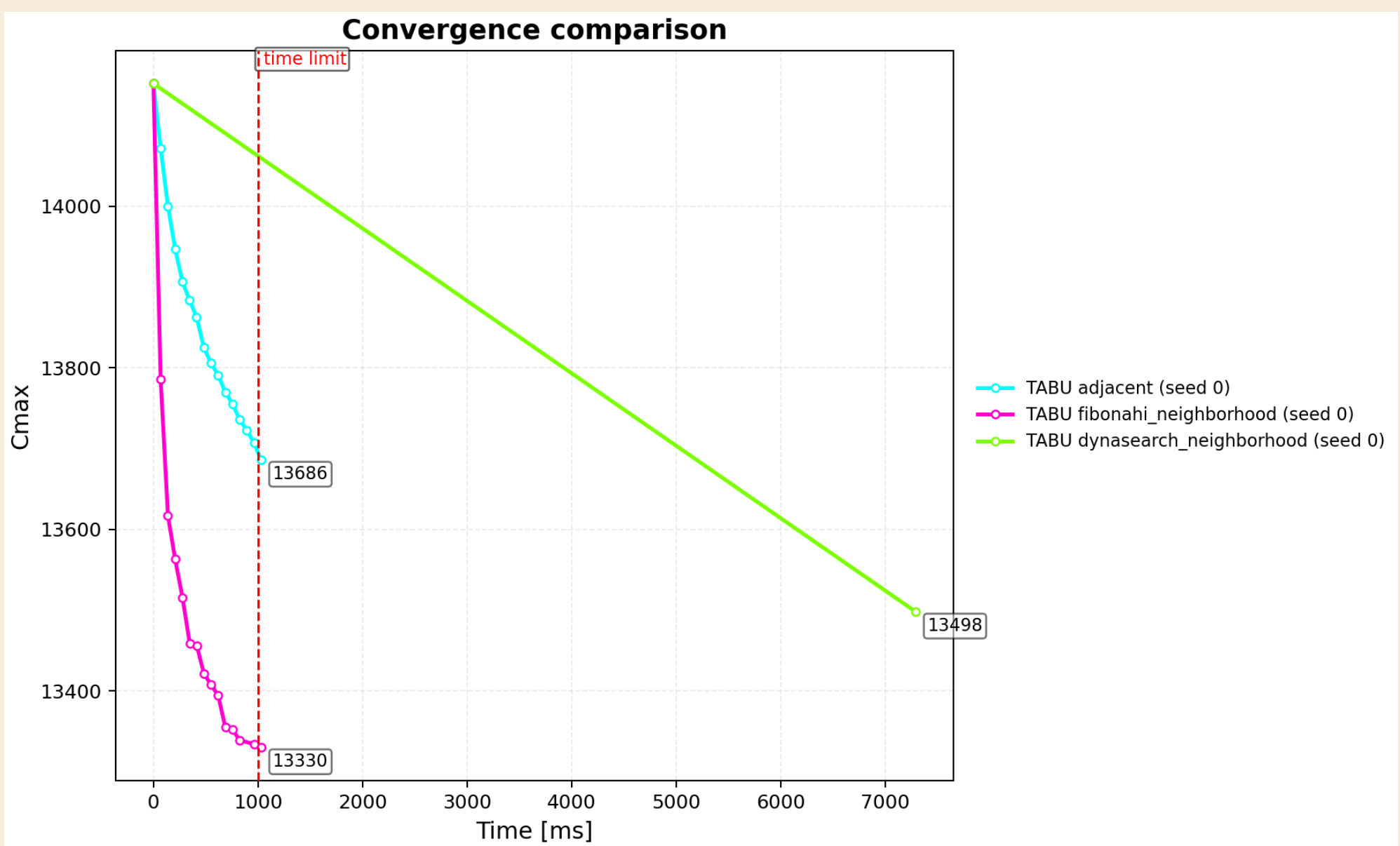
- Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$ — 12 instance classes.
- Time limits: $tl \in \{100, 500, 1000, 2000, 5000, 10000\}$ ms.
- SA: $T_0 = 1000$, $T_{\text{final}} = 1$, $\alpha = 0.95$, reheat $r = 1.5$, stagnation 500 ms.
- Platform: MacBook Pro M4 Pro, 24 GB RAM, Python 3.11.
- Metric: PRD to best-known $C_{\max}^*$:
 $\text{PRD} = \frac{C_{\max} - C_{\max}^*}{C_{\max}^*} \cdot 100\%.$

6. Results — PRD at $tl = 5,000$ ms

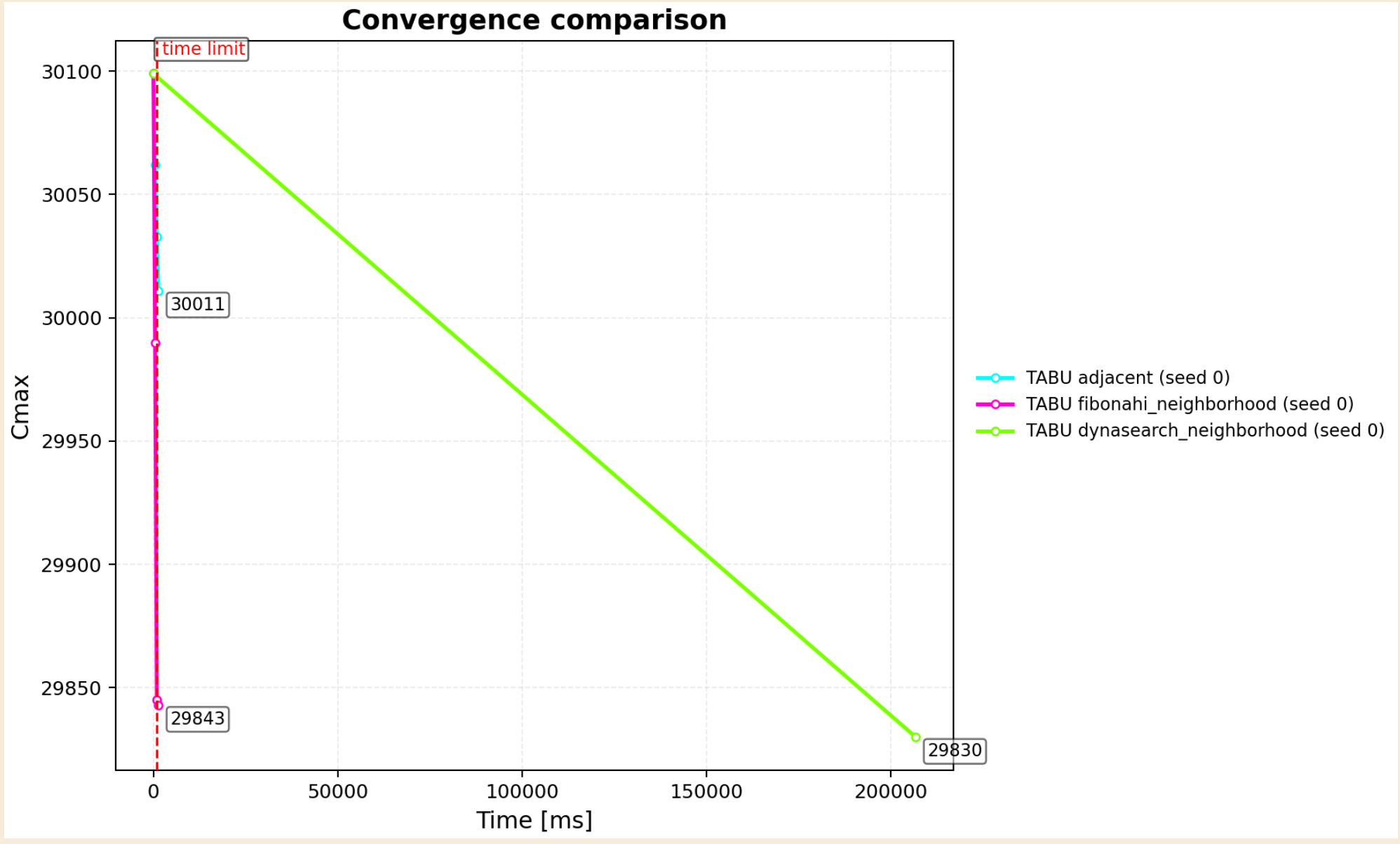
instance	n	m	fib _{TS}	fib _{SA}	dyn _{TS}	dyn _{SA}	adj _{TS}	adj _{SA}
tai20_5	20	5	17.39	17.39	6.18	6.18	4.17	6.24
tai50_10	50	10	20.23	20.23	6.95	6.95	17.10	9.72
tai100_10	100	10	17.19	17.19	6.57	6.48	16.52	8.27
tai200_10	200	10	12.71	12.71	13.20	13.20	13.19	8.53
tai200_20	200	20	18.60	18.60	20.71	20.71	19.83	20.20
tai500_20	500	20	13.91	13.90	14.51	14.51	15.34	15.82

Crossover at $n = 200$: *Fibonacci* **overtakes** both Adjacent and Dynasearch and remains competitive at $n = 500$, where Dynasearch cannot complete a single iteration within the budget.

7. Convergence — $tl = 1,000$ ms



tai200_20, $n = 200$, $m = 20$



tai500_20, $n = 500$, $m = 20$

Fibonacci (magenta) drops $C_{\max}$ fastest in the first ~ 100 ms; Dynasearch (orange) barely completes one iteration within $tl = 1$ s; Adjacent (cyan) makes slow, steady progress.

8. Discussion

- Early-budget gain.** Composite move bundles several non-overlapping improving swaps into one iteration.
- Plateau risk.** Once all $\Delta_i \geq 0$, the composite degenerates to a single swap; SA reheating mitigates this.
- Dynasearch cost.** A single iteration at $n = 500$ can exceed tl by $\sim 200\times$.
- Practical takeaway.** *Fibonacci* is the right default for medium-to-large n under tight time limits; Adjacent for very small n ; Dynasearch only when ample budget is available.

9. Conclusion & future work

Summary. Three neighborhoods compared inside TS and SA on Taillard. *Fibonacci* (Composite Disjoint-Adjacent) dominates from $n = 200$ onward; Dynasearch reaches deeper optima but is impractical for large n within short time budgets.

Future work.

- Incremental delta evaluation for the composite move ($O(mn^2) \rightarrow O(mn)$).
- Adaptive switching between neighborhoods at runtime.
- Comparison with state-of-the-art FSP solvers (iterated LS, VND).
- Quantum-assisted Fibonacci selection on D-Wave (windowed QUBO, n up to 500).

Acknowledgements

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